MATH 301: Homework 1

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Question 1: From Blackboard

Proposition: Let S and T be two sets. If both S and T are countable, then $S \cup T$ is countable.

Proof. Suppose S and T are countable, and there exists $A, B \subseteq \mathbb{N}$, by definition there exists a bijective function:

$$f : A \to S \text{ and } g : B \to (T - S)$$

Observe we can construct $S \cup T$ as a disjoint set. Let S, T be defined as follows:

$$S = \{s | s \in S\}, T = \{t | t \in T\}$$

Natrually then.

$$S \cup T = \{x | x \in S \vee x \in T\}$$

Observe:

$$S \cup T - S = \{y | y \in S \vee (y \in T \wedge y \notin S)\}$$

Consider the following statments (note reversible):

$$\begin{aligned}
&\iff S \vee (y \in T \wedge y \notin S) \\
&\iff S \vee (y \in T \wedge y \in S^c) \\
&\iff y \in S \cup (T \cap S^c) \\
&\iff y \in (S \cup T) \cap (S \cup S^c) \\
&\iff y \in (S \cup T) \cap U \\
&\iff y \in S \cup T
\end{aligned}$$

Thus we can state that $S \cup T = S \cup T - S$.

To show $S \cup T$ is countable, we must construct a bijection $h : C \to S \cup (T - S), C \subseteq \mathbb{N}$
Define $h : C \to S \cup (T - S)$:

$$h(n) = \begin{cases} f\left(\frac{n}{2}\right) & \text{if } n \text{ is even} \\ g\left(\frac{n+1}{2}\right) & \text{if } n \text{ is odd} \end{cases}$$

> **Aside:** $S, (T - S)T$ must be disjoint.
> $\exists\ n_1, n_2 \in C, n_1 \neq n_2 \ni h(n_1) = f(n_1) = h(n_2) = g(n_2)$

To prove the proposition we must show that h is *surjective* and *injective.*

Surjectivity

Goal: We want to show that $\forall\ y \in S \cup T, \exists\ n \in C \ni h(n) = y$

Case 1 ($y \in S$):

Let $y \in S$. Since f is surjective, $\exists\ k \in C$ s.t. $f(k) = y$.
Let $n = 2k$. By definition n is even. Observe:

$$h(n) = f(\frac{2k}{2}) = f(k) = y$$

Case 2 $(y \in T)$**:**

Let $y \in T$. Since g is surjective, $\exists\ k \in C \ni g(k) = y$
Let $n = 2k - 1$. Then n is odd. Observe:

$$h(n) = g(\frac{(2k-1)+1}{2}) = g(\frac{2k}{2}) = g(k) = y$$

> **Recall:** *Countable Set Implications*
> The following statements are equivalent:
>
> $$S \text{ is countable.} \iff \exists \text{ surjection of } \mathbb{N} \text{ onto } S. \iff \exists \text{ an injection of } S \text{ into } \mathbb{N}.$$

$$\text{As every element } y \in S \cup T\ \exists\ n \in C \ni h(n) = y \implies h : C \to S \cup (T - S) \text{ is surjective.}$$

Note since $h(n) : C \to S \cup (T - S)$ is surjective and $C \subseteq \mathbb{N}$.
$\implies$ Thus we can validate a condition in "Countable Set Implications" ($\exists$ surjection $\mathbb{N}$ onto S).
$\implies$ It means there must be an injection of $S \cup (T - S) \to \mathbb{N}$.

$$\text{This means that } S \cup (T - S) \text{ must be countable.}$$

□

Question 2: Sec 1.3 (q1)

Question: Prove that a nonempty set T_1 is finite iff $\exists$ a bijection from T_1 onto a finite set T_2.

Case 1 $\to$ **:**

Note: To prove this by "Pigeon Hole Principle" we must suppose $|T_1| = |T_2|$ else there does not exist a way to construct a bijection.

Proof. (**Show:** If T_1 is finite. Then $\exists$ a $f : T_1 \to T_2$, where T_2 is finite and f is bijective)
Suppose T_1 is finite. Consider a function $f : T_1 \to T_2$ defined as:

$$f := \{(a_i, b_j) | a_i \in T_1 \text{ and } b_j \in T_2\}$$

Let a_i be an arbitrary element defined to be (where all elments are distict):

$$a_i = a_1, a_2, a_3, ..., a_{|T_1|},$$

Let b_i be an arbitrary element defined to be (where all elments are distict):

$$b_i = b_1, b_2, b_3, ..., b_{|T_2|},$$

Clearly:

$$\forall\ b \in T_2, \exists\ a \in T_1 \ni f(a) = b$$

Thus f is surjective.

Clearly:

$$f(x) = f(y) \implies x = y$$

Thus f is injective.

Conclusion: As f is both injective and surjective, given T_1 is finite then clearly there exists a bijection from T_1 onto T_2. □

Case 2 $\leftarrow$:

Proof. (**Show:** If $\exists$ a $f : T_1 \to T_2$, where T_2 is finite and f is bijective. Then T_1 is finite.)
Suppose $\exists$ a $f : T_1 \to T_2$, where T_2 is finite and f is bijective.
For the sake of contradiction supose T_1 is infinite.
As f is bijective we can contruct an arbitrary mapping as:

$$f := \{(a_i, b_j) | a_i \in T_1 \text{ and } b_j \in T_2\}$$

Let a_i be an arbitrary element defined to be (where all elments are distict):

$$a_i = a_1, a_2, a_3, ..., a_{|T_1|},$$

Let b_i be an arbitrary element defined to be (where all elments are distict):

$$b_i = b_1, b_2, b_3, ..., b_{|T_2|},$$

However as T_1 is infinite and T_2 is finite then clearly $|T_2| < |T_1|$.
Thus for the function $f : T_1 \to T_2$ to be a function the following must be true (by "Pigeon Hole Principle"):

$$f(x) = f(y) \implies x \neq y$$

Conclusion: This is a contradiction, thus we can say: if $\exists$ a $f : T_1 \to T_2$, where T_2 is finite and f is bijective. Then T_1 is finite. □

As we have proved both Case 1 and Case 2 we have shown that a nonempty set T_1 is finite iff $\exists$ a bijection from T_1 onto a finite set T_2.

Question 3: Sec 1.3 (q4)

Proof. **Question:** Exhibit a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13.

Setup

We want to find a set of *all odd integers greater than 13*. Let's define this as S (where $i \in \mathbb{N}$):

$$S = \{15, 17, 18, ...i\} = \{2z + 13 | z \in \mathbb{Z}\}$$

(*Note: all elements defined are distict*)

Also let's define elements of $\mathbb{N}$ as follows, $j \in \mathbb{N}$:

$$n_1 = 1$$
$$n_2 = 2$$
$$n_3 = 3$$
$$...$$
$$n_j = j$$

(*Note: all elements defined are distict*)

Lets construct a function that maps $\mathbb{N}$ to S:

$$f : \mathbb{N} \to S = \{(s_i, n_j) | s_i \in S \wedge n_j \in \mathbb{N}\}$$

Observe this is modeling the function as:

$$f(n) = 13 + 2n$$

Surjectivity

Proof. **Goal:** Show that $\forall\ s \in S, \exists\ n \in \mathbb{N} \ni f(n) = s$
Let $s \in S, n \in \mathbb{N}$:

$$s_n = 13 + 2n = f(n)$$

By algebra:

$$\begin{aligned} s_n &= 13 + 2n \\ 2n &= s_n - 13 \end{aligned}$$

Recall $\forall\ s \in S$, s must be an odd integer greater than 13. Thus there must be some $k \in \mathbb{N}$ s.t.

$$s = 2k + 1$$

By subsitution:

$$\begin{aligned} 2n &= s_n - 13 \\ 2n &= (2k + 1) - 13 \\ 2n &= 2k + (1 - 13) \\ 2n &= 2k - 12 \\ n &= k - 6 \end{aligned}$$

Now we must ask if n as defined to be as $n = k - 6$ escapes the set $\mathbb{N}$ as n is defined to be $n \in \mathbb{N}$. As s is defined to be *greater than 13*. We know there exists a greatest lower bound 15 that satisfies the criterion. By subsitution:

$$\begin{aligned} s &> 13 \\ s &= 2k + 1 \\ 2k + 1 &> 13 \\ 2k + 1 - 1 &> 13 - 1 \\ 2k &> 12 \\ k &> 6 \end{aligned}$$

As $k > 6$ then this means the $n = k - 6$ must always be greater than zero. As all operations retain closure properties of integers, we can expect the value to be an interger. By def all integers greater than zero must belong to $\mathbb{N}$. Thus we can say $n \in \mathbb{N}$. Thus the properties of surjectivity hold. **Backsolve:**

$$\begin{aligned} f(k-6) &= 13 + 2(k - 6) \\ f(k-6) &= 13 + 2k - 12 \\ f(k-6) &= 13 - 12 + 2k \\ f(k-6) &= 1 + 2k = s \end{aligned}$$

We have shown that $\forall\ s \in S, \exists\ n \in \mathbb{N} \ni f(n) = s$

□

Injectivity

Proof. **Goal:** Show that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$
Let $a, b \in \mathbb{N}$. Suppose:

$$f(a) = f(b)$$

By subsitution:

$$\begin{aligned} 2a + 1 &= 2b + 1 \\ 2a + 1 - 1 &= 2b + 1 - 1 \\ 2a &= 2b \\ a &= b \end{aligned}$$

$$\text{Thus we have shown that given } a, b \in \mathbb{N} \text{ and } f(a) = f(b) \implies a = b$$

□

Conclusion

As f is both injective and surjective we have shown there exists a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13. □

Question 4: Sec 1.3 (q6)

Question: Exhibit a bijection between $\mathbb{N}$ and a proper subset of itself.
Let's construct a set E, a set of all even natural numbers, and let $k \in \mathbb{N}$:

$$E := \{n | n = 2k\}$$

$$\text{Clearly } 1 \in \mathbb{N} \text{ and } 1 \notin E \text{ as 1 is not even. Therefore } E \subset \mathbb{N}.$$

To prove a bijection exists, we must construct a function $f : \mathbb{N} \to E$ that is surjective and injective. Let's construct the function as follows, let $n \in \mathbb{N}, v \in E$:

$$f(n) = 2n = v$$

Surjectivity

Proof. **Goal:** Show that $\forall\ s \in S, \exists\ n \in \mathbb{N} \ni f(n) = s$
Let $v \in E, n \in \mathbb{N}$. We can say the following:

$$v = 2n \implies \frac{v}{2} = n$$

Recall that $v \in E$, which constricts the elements of E to be even, as defined. Let $k \in \mathbb{N}$ and $v = 2k$. Thus we can rearrange the equation as follows:

$$\frac{v}{2} = n \implies \frac{2k}{2} = n \implies k = n$$

Now notice we have not violated any closure properties thus the solution remains valid.

Backsolve: to confirm validity.

$$f(k) = 2k = v$$

$$\textit{Recall: we had defined } v = 2k\textit{, so this is the expected answer.}$$

Conclusion:

$$\text{Thus, we have show that } \forall\ s \in S, \exists\ n \in \mathbb{N} \ni f(n) = s$$
$$\text{Therefore } f \text{ is surjective.}$$

$\square$

Injectivity

Proof. **Goal:** Show that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$
Let $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$. Observe:

$$\begin{aligned} f(a) &= f(b) \\ 2a &= 2b \\ \frac{2a}{2} &= \frac{2b}{2} \\ a &= b \end{aligned}$$

$$\text{Thus we have shown that given } a, b \in \mathbb{N} \text{ and } f(a) = f(b) \implies a = b$$
$$\text{Therefore } f \text{ is injective.}$$

$\square$

Conclusion

As f is both injective and surjective we have shown there exists a bijection between $\mathbb{N}$ and the set of all even natural numbers.

Question 5: Sec 1.3 (q7)

Question: Prove that a set T_1 is denumerable iff there is a bijection from T_1 onto a denumerable set T_2

Case 1 $\rightarrow$:

Prove: Given a denumerable set T_1 we can construct a bijection onto a denumerable set T_2.

Proof. Suppose T_1 is denumerable. This implies a bijective function f exists:

$$f : \mathbb{N} \rightarrow T_1$$

Consider the existance of another denumerable set T_2. Likewise a bijective fuction g must also exist:

$$g : \mathbb{N} \rightarrow T_2$$

As these functions are bijective, this implies the existance of an inverse function for each.

$$f^{-1} : T_1 \rightarrow \mathbb{N} \text{ and } g^{-1} : T_2 \rightarrow \mathbb{N}$$

Goal: we want to show there exists a function $h : T_1 \to T_2$ that is bijective.
Approach: consider the change in space

$$T_1 \to \mathbb{N} \to T_2$$

Consider the composition function:

$$g \circ f^{-1} : T_1 \to T_2$$

From definition above we know $Img(f^{-1}) = \mathbb{N}$, via the implication of bijection.
We also know that $g : \mathbb{N} \to T_2$ is bijective.
Thus we can say the following is a valid bijective function:

$$g : Img(f^{-1}) \to T_2 \equiv g : \mathbb{N} \to T_2$$

Conclusion: Thus we have shown given a denumerable set T_1 we can construct a bijection towards another denumerable set T_2.

□

Case 2 $\leftarrow$:

Prove: Given a denumerable set T_2, if a bijection exists from T_1 onto T_2 then T_1 must be denumerable.

Proof. Suppose T_2 is a denumerable set. Thus, there must exist a bijection from T_2 and $\mathbb{N}$. Thus there must exist some functions:

$$f : \mathbb{N} \to T_2 \text{ and } f^{-1} : T_2 \to \mathbb{N}$$

Suppose there exists a bijection from T_1 onto T_2. We can define the following functions:

$$g : T_1 \to T_2 \text{ and } g^{-1} : T_2 \to T_1$$

Approach: consider the change in space

$$T_1 \to T_2 \to \mathbb{N}$$

We can construct the composition:

$$f^{-1} \circ g : T_1 \to \mathbb{N}$$

Consider the image set:

$$Img(g) = T_2$$

By subsitution:

$$f^{-1}(Img(g)) = \mathbb{N}$$

As these functions are bijective, we can consider their inverses.

$$f(\mathbb{N}) = T_2 \implies g(T_2) = T_1 \implies g(f(\mathbb{N})) = T_1$$

This means there exists a bijective function from $\mathbb{N} \to T_1$. Therefore, T_1 must be denumerable.
Conclusion: We have shown given a denumerable set T_2, if a bijection exists from T_1 onto T_2 then T_1 must be denumerable. □

Question 6: Sec 1.3 (q8)

Question: Give an example of a countable collection of finite sets whose union is not finite.
Consider the sets defined to be:

$$A_n = \{n\}, \forall\ n \in \mathbb{N}$$

These sets are finite sets defined as:

$$\begin{aligned} &A_1 = \{1\} \\ &A_2 = \{2\} \\ &A_3 = \{3\} \\ &... \\ &A_n = \{n\} \end{aligned}$$

Observe each set $A_1, A_2, A_3, ..., A_2$ are finite (singleton) sets.

Though the union of all sets, by how we defined $A_1, A_2, A_3, ..., A_2$, would be result in the equivalent set $\mathbb{N}$ which is not finite.